

Name of the Student	Narra Sai Mounika
Reg. No	192425367
GitHub Link	https://github.com/saimounika1315/MLA0403---Deep-Learning

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning
SLOT : A

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%

Assessment Tool Weightage: 40%

Duration: 90 minutes

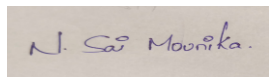
Mark : 25

Code of Conduct:

I **Narra Sai Mounika / 192425367** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Narra Sai Mounika



Signature of the student:

Name of the student:

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

QUESTIONS: 05

Total Marks:25

Q1. Transfer Learning

A healthcare startup aims to develop an AI-based system for detecting skin cancer from dermoscopic images. Since only a limited number of labeled images are available, design a transfer learning strategy using a pre-trained CNN model. Justify your choice of model, identify which layers should be frozen or fine-tuned, and explain how your design improves classification performance.

Answer:

Understanding the Problem

The healthcare startup has a limited number of dermoscopic images. Training a CNN from scratch may lead to overfitting. Transfer learning helps utilize knowledge from large image datasets.

Proposed Architecture

Dermoscopic Image



Pre-trained ResNet50
(Frozen Initial Layers)



Global Average Pooling



Dense Layer (256, ReLU)



Dropout (0.5)



Output Layer (Benign / Malignant)

Design Strategy

- Use **ResNet50** pre-trained on ImageNet.
- Freeze the initial convolutional and residual layers.
- Fine-tune the last residual block.
- Replace the original classification layer with:
 - Dense (256, ReLU)
 - Dropout (0.5)
 - Softmax Output Layer

Tools Used

- TensorFlow/Keras
- ResNet50 Pre-trained Model
- Data Augmentation (rotation, zoom, flip)

Justification

- Reduces training time.
- Prevents overfitting on small datasets.
- Learns medical image features efficiently.
- Improves skin cancer classification accuracy.

Analysis

Transfer learning enables the model to achieve high accuracy even with limited labeled images while reducing computational cost.

Q2. DenseNet and PixelNet

A smart city project requires an automated road-scene analysis system that accurately identifies roads, pedestrians, vehicles, and traffic signs at the pixel level. Design a deep learning architecture using DenseNet and PixelNet to solve this problem. Explain how the proposed architecture improves feature reuse, segmentation accuracy, and computational efficiency.

Answer:

Understanding the Problem

The system must classify every pixel into:

- Road
- Pedestrian
- Vehicle
- Traffic Sign

This is a semantic segmentation problem.

Proposed Architecture

Input Road Image



DenseNet Backbone



Dense Feature Maps



PixelNet Segmentation Module



Pixel-wise Classification



Segmented Output Image

Role of DenseNet

DenseNet connects each layer to all previous layers.

Layer1 → Layer2

Layer1 → Layer3

Layer2 → Layer3

Layer1 → Layer4

Layer2 → Layer4

Layer3 → Layer4

Role of PixelNet

- Combines low-level and high-level features.
- Produces pixel-level predictions.
- Improves boundary detection.

Advantages

Feature Reuse

- DenseNet reuses extracted features.
- Reduces redundant computations.

Segmentation Accuracy

- PixelNet captures detailed object boundaries.
- Improves road and pedestrian detection.

Computational Efficiency

- Fewer parameters.
- Better gradient flow.

- Faster convergence.

Tools Used

- DenseNet121
- PixelNet
- TensorFlow/PyTorch
- Cityscapes Dataset

Analysis

The DenseNet-PixelNet combination provides accurate semantic segmentation with efficient feature reuse and lower computational complexity.

Q3. Bidirectional RNN and Encoder–Decoder

A multinational customer service company wants to build a multilingual chatbot capable of translating customer queries while preserving contextual meaning. Design an Encoder–Decoder sequence-to-sequence model using Bidirectional RNNs. Illustrate the architecture and justify how bidirectional processing improves translation accuracy.

Answer:

Understanding the Problem

The chatbot must translate customer queries while preserving contextual meaning.

Proposed Architecture

Input Sentence



Word Embedding



Bidirectional RNN Encoder



Context Vector



Decoder RNN



Softmax Layer



Translated Sentence

Bidirectional Encoder

Forward Pass:

Word1 → Word2 → Word3 → Word4

Backward Pass:

Word4 → Word3 → Word2 → Word1

Combined Hidden State:

Forward State + Backward State

Working Process

Encoder

- Reads complete source sentence.
- Captures contextual information.

Context Vector

- Stores semantic representation.

Decoder

- Generates translated sentence word by word.

Why Bidirectional RNN?

Example:

The bank is near the river.

The meaning of "bank" depends on future words.

Bidirectional processing captures:

- Previous context
- Future context

Tools Used

- TensorFlow/Keras
- Bidirectional LSTM/GRU
- Seq2Seq Framework

Analysis

Bidirectional RNNs improve translation quality by understanding complete sentence context, resulting in more accurate multilingual communication.

Q4. BPTT and LSTM

A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

Ans:

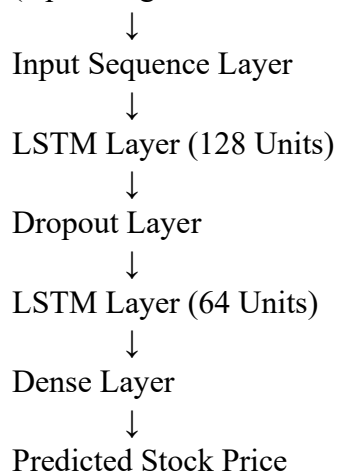
Understanding the Problem

Stock prices are sequential and depend on historical market trends.

Proposed Architecture

Historical Stock Data

(Open, High, Low, Close, Volume)



Training using BPTT

Forward Pass

- Historical data passes through LSTM.
- Predictions are generated.

Error Calculation

Error = Actual Price - Predicted Price

Backpropagation Through Time (BPTT)

Time1 ← Time2 ← Time3 ← Time4

- Error propagates backward through all time steps.
- Weights are updated using gradient descent.

Why LSTM over Traditional RNN?

Traditional RNN

Suffers from vanishing gradient
Poor memory retention
Less accurate for long sequences

LSTM

Handles long-term dependencies
Memory cell stores information
Better forecasting accuracy

Tools Used

- TensorFlow/Keras
- Yahoo Finance Dataset
- Adam Optimizer

Analysis

LSTM effectively captures long-term market trends, while BPTT enables efficient learning from sequential stock data.

Q5. Integrated Deep Learning Solution

A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for caption generation. Explain the role of each component and justify how the complete system produces accurate captions.

Ans:

Understanding the Problem

The system must automatically generate captions for uploaded videos.

Proposed Architecture

Video Frames



Transfer Learning CNN
(ResNet50/InceptionV3)



Visual Feature Vectors



Encoder LSTM



Context Vector



Decoder LSTM



Softmax Layer



Generated Caption

Component Roles

1. Transfer Learning CNN

- Extracts visual features from video frames.
- Uses pre-trained ResNet50/InceptionV3.
- Converts images into feature vectors.

Example:

Frame → CNN → Feature Vector

2. Encoder LSTM

- Processes sequence of frame features.

- Learns temporal relationships.

3. Context Vector

- Stores summarized video information.

4. Decoder LSTM

- Generates caption word-by-word.

Example:

"A man is riding a bicycle."

5. Softmax Output Layer

- Predicts the most probable next word.

Training Process

1. Extract video frames.
2. Generate CNN features.
3. Feed features to Encoder LSTM.
4. Train Decoder using caption datasets.
5. Optimize using cross-entropy loss.

Tools Used

- ResNet50 / InceptionV3
- LSTM Encoder–Decoder
- TensorFlow/Keras
- MSR-VTT or MSVD Dataset

Analysis

The transfer learning CNN captures rich visual information, while the Encoder–Decoder LSTM learns temporal context and language structure. This combination generates accurate, meaningful, and grammatically correct captions, improving video accessibility and user experience

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis